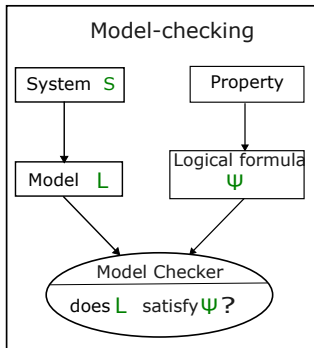


A Theory of (Linear-Time) Timed Monitors

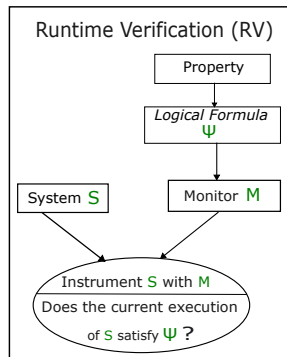
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Model-checking vs. Runtime Verification



- ▶ Exhaustive.
- ▶ A model of the system is needed.
- ▶ State-space explosion.



- ▶ Suitable for black-box systems.
- ▶ Scalable.
- ▶ Non-exhaustive.
- ▶ Monitorability.

Monitorability in a timed setting

We consider both the sequence of actions and the delays between them.

Example (The 'bounded response' property)

Every action **b** is eventually followed by an action **c** within 1 time unit (relative to the occurrence of **b**).

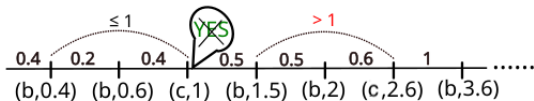


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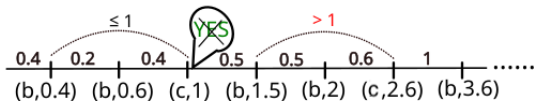


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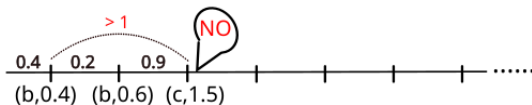
This property is not monitorable for satisfactions!

Monitorability in a timed setting

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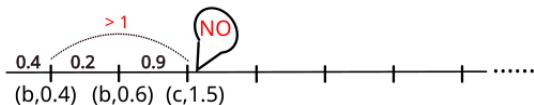


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We consider both the sequence of actions and the delays between them.

Example (The 'bounded response' property)

Every action **b** is eventually followed by an action **c** within 1 time unit (relative to the occurrence of **b**).



This property is monitorable for violations...

The Monitorability problem in a timed setting

- ▶ In a nutshell, the “game” of monitorability is about telling the future.
- ▶ Given some formula ϕ , can I correctly issue, after a finite observation, a :
 - ▶ *Correct* satisfaction verdict for every infinite execution satisfying ϕ (satisfaction monitorability),
 - ▶ *Correct* violation verdict for every infinite execution violating ϕ (violation monitorability).

Problem

Given an expressive timed logic, which formulae are monitorable?

The monitorability problem has been solved **only** for *untimed* logics, notably the untimed modal μ -calculus.

Real-time executions

- ▶ Given a finite set of actions ACT , a silent action τ , we define the set of labels $\text{LAB} = \text{ACT} \cup \{\tau\} \cup \mathbb{R}_{\geq 0}$.
- ▶ A *real-time system* is a timed labelled transition system (TLTS) over LAB .
- ▶ A real-time execution (run) starting from a program (state) p_1 is an infinite sequence of transitions in the TLTS.
 - ▶ for example

$$p_1 \xrightarrow{0.5} p_2 \xrightarrow{\tau} p_3 \xrightarrow{0.3} p_4 \xrightarrow{b} p_5 \xrightarrow{1.2} p_6 \xrightarrow{c} p_7 \dots$$

- ▶ $0.5 \ \tau \ 0.3 \ b \ 1.2 \ c \ \dots$ is a run (in LAB^ω) starting from p_1 .
- ▶ Let $\mathcal{L} = \text{LAB} \setminus \{\tau\}$. A finite weak (observable) run in $\mathcal{L}^*$ is obtained from a prefix of a run
 - ▶ example $0.8 \ b \ 1.2 \ c$ from $0.5 \ \tau \ 0.3 \ b \ 1.2 \ c$
- ▶ for some states p, p' and weak finite run $\theta \in \mathcal{L}^*$, we write:
 - ▶ $p \xrightarrow{\theta} p'$ to denote that p' is reachable from p along θ ,

Real-time executions

- ▶ An *infinite-timed-word* (trace) representation is used to evaluate logical formulae
- ▶ Let $\text{TACT} = \text{ACT} \times \mathbb{R}_{\geq 0}$. A trace is of the form $(a_1, t_1)(a_2, t_2) \dots$ in TACT^ω such that $t_i \leq t_{i+1}$ for every $i \in \mathbb{N}_{>0}$.
 - ▶ Here the *timestamp* t_i represents the *absolute time* at which action a_i occurred.
 - ▶ A finite trace in TACT^* is a prefix of a trace.
 - ▶ Each finite trace *corresponds* therefore to a unique finite weak run using $\text{TRN} : \text{TACT}^* \longrightarrow \mathcal{L}^*$:

$$\text{TRN}((a_1, t_1)(a_2, t_2) \dots (a_n, t_n)) = d_1 \ a_1 \ d_2 \ a_2 \dots \ d_n \ a_n$$

$$\text{with } d_i = t_i - t_{i-1} \text{ and } t_0 = 0$$

- ▶ Example: $\text{TRN}((a, 1.7)(b, 1.7)(b, 5.7)) = 1.7 \ a \ 0 \ b \ 4 \ b$

Our timed mu-calculus T^{lin}

$\phi, \psi \in T^{lin}$	$::=$	tt	(true)
		ff	(false)
		X	(logical variable)
		$\phi \wedge \psi$	(conjunction)
		$\phi \vee \psi$	(disjunction)
		$\langle a \rangle \phi$	(diamond)
		$\langle g \rangle \phi$	(diamond on time)
		$x \text{ in } \phi$	(time-stamp recording)
		$\min(X, \phi)$	(min. fixed-point)
		$\max(X, \phi)$	(max. fixed-point)

$$g ::= x \prec n \quad | \quad (x - y) \prec n \quad | \quad n \prec x \quad | \quad n \prec (x - y) \quad | \quad g \wedge g \quad | \quad \neg g$$

- ▶ X is a **recursion variable**, x, y are **timestamp variables**, n is an integer.
- ▶ A **formula** is a **closed** term generated by the grammar above.
- ▶ Notation shortcuts: given non empty $A \subseteq \text{ACT}$, $\langle A \rangle \phi = \bigvee_{a \in A} \langle a \rangle \phi$, $\bar{a} = \text{ACT} \setminus \{a\}$, $\bar{g} = \neg g$.
- ▶ syntactic sugar: $[g]\phi = \langle g \rangle \phi \vee \langle \bar{g} \rangle \text{tt}$, $[a]\phi = \langle a \rangle \phi \vee \langle \bar{a} \rangle \text{tt}$.

Semantics of T^{lin} : a (simple) example

Example

The property 'The first action is *start* and occurs within 3 time units' is expressed by the T^{lin} formula:

$$x \text{ \underline{in} } \langle x \leq 3 \rangle \langle \textit{start} \rangle \text{tt}$$

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$$\pi = (start, 1.5) \dots$$

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$$\pi = (\text{start}, 1.5) \dots \in \llbracket x \underline{\text{in}} \langle x \leq 3 \rangle \langle \text{start} \rangle \text{tt} \rrbracket$$

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$$\pi' = (\textcolor{red}{b}, \textcolor{green}{1.5}) \dots \notin \llbracket x \underline{\text{in}} \langle x \leq 3 \rangle \langle \text{start} \rangle \text{tt} \rrbracket$$

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$$\pi'' = (b, 3.4) \dots \notin \llbracket x \text{ \underline{in} } \langle x \leq 3 \rangle \langle \textit{start} \rangle \texttt{tt} \rrbracket$$

Semantics of T^{lin} : a (not-so-simple) example

- ▶ Let $\mathcal{I}$ be the set of non-empty intervals over $\mathbb{R}_{\geq 0}$ with a left bound in $\mathbb{N}$, and a right bound in $\mathbb{N} \cup \{+\infty\}$.
- ▶ Let $\phi_{even(b),I} \in T^{lin}$, parameterised with $I \in \mathcal{I}$, a formula defined as follows

$$x \text{ \underline{in} } \overbrace{\max(X, \langle \text{ACT} \rangle (\underbrace{y \text{ \underline{in} } \langle y - x \in I \rangle \langle \mathbf{b} \rangle X}_{\psi_1}) \vee (\underbrace{z \text{ \underline{in} } \langle z - x \notin I \rangle \langle \text{ACT} \rangle X}_{\psi_2}))}_{\psi}$$

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- ▶ π is in $\llbracket \phi_{even(b),I} \rrbracket$ iff at every even position of π , if the action occurs within I then it is a **b**.
- ▶ Example for $\pi = (c, 1)(b, 3.5)(b, 5)(c, 8) \dots$ is in $\llbracket \phi_{even(b),[2,6]} \rrbracket$,
- ▶ $\pi' = (c, 0)(b, 3.5)(b, 5)(c, 5.5)(b, 8) \dots$ is **not** in $\llbracket \phi_{even(b),[2,6]} \rrbracket$.

Expressiveness

Theorem

T^{lin} is strictly more expressive than Metric Temporal Logic (MTL):

1. For every $\phi \in MTL$, there exists $\psi \in T^{lin}$ such that $\llbracket \phi \rrbracket = \llbracket \psi \rrbracket$
2. There exists $\psi \in T^{lin}$ such that for every $\phi \in MTL$ we have $\llbracket \phi \rrbracket \neq \llbracket \psi \rrbracket$

► To prove item 1, we propose $TO T^{lin} : MTL \longrightarrow T^{lin}$ such that:

- For every $\phi \in MTL$ we have $\llbracket \phi \rrbracket = \llbracket TO T^{lin}(\phi) \rrbracket$.
- example

$$TO T^{lin}(\mathbf{b} \mathcal{U}_I \mathbf{c}) = x_s \underline{\text{in}} \overbrace{\min(X, \underbrace{(x_e \underline{\text{in}} \langle x_e - x_s \in I \rangle \langle \mathbf{c} \rangle \mathbf{tt})}_{\psi_1} \vee \underbrace{(\langle \mathbf{b} \rangle \mathbf{tt} \wedge \langle \text{ACT} \rangle X)}_{\psi_2})}_{\psi}$$

► for item 2, $\phi_{\text{even}(b), I}$ has no equivalent counterpart in MTL

Monitors

$M, N \in \text{MON}$	$::= E$	$ a.M$	$ X$
	$ g.M$	$ M + N$	$ \text{rec } X.M$
	$ \text{LET } x \text{ IN } M$	$ M \otimes N$	$ M \oplus N$
E	$::= \text{VYES}$	$ \text{VNO}$	$ \text{VEND}$
VYES	$::= \text{YES}$	$ \text{VYES} \oplus M$	$ \text{VYES} \otimes \text{VYES}$
VNO	$::= \text{NO}$	$ \text{VNO} \otimes M$	$ \text{VNO} \oplus \text{VNO}$
VEND	$::= \text{END}$	$ \text{VEND} \oplus \text{VEND}$	$ \text{VEND} \otimes \text{VEND}$

Instrumentation

- ▶ $\mathbb{M}$ is the TLTS representing the semantics of all monitors in MON (examples will follow).
 - ▶ A monitor state is a triplet (M, v_c, v_t) where $M \in \text{MON}$, $v_c \in \mathbb{R}_{\geq 0}$ is a global “clock” denoting the absolute time and v_t is a valuation of timestamp variables.
 - ▶ yes, no, end are shorthands for verdict states.
- ▶ A monitor is a *passive entity*, it is *at the mercy* of the monitored program.

$$\frac{p \xrightarrow{\alpha}_P p' \quad m \xrightarrow{\alpha}_{\mathbb{M}} m' \quad \alpha \in \text{TACT}}{m \triangleleft p \xrightarrow{\alpha}_{\triangleleft} m' \triangleleft p'} \text{IMON} \qquad \frac{p \xrightarrow{\tau}_P p'}{m \triangleleft p \xrightarrow{\tau}_{\triangleleft} m \triangleleft p'} \text{IASY}$$

$$\frac{p \xrightarrow{a}_P p' \quad m \xrightarrow{a}_{\mathbb{M}} \quad m = (N, v_c, v_t)}{m \triangleleft p \xrightarrow{a}_{\triangleleft} (\text{END}, v_c, v_t) \triangleleft p'} \text{ITER}$$

Instrumentation : example

Let

$$M = \text{LET } x \text{ IN } \underbrace{\text{rec } X.((x \in [1, 3])(\mathbf{b}.X + \mathbf{c}.\text{NO}) + (x \notin [1, 3]).\text{YES}))}_N$$

and $p \xrightarrow{1.6} p_1 \xrightarrow{b} p_2 \xrightarrow{c} p_3 \xrightarrow{4} p_4 \xrightarrow{b} p_5 \dots$. Then we have

$$m_0 \triangleleft p \xrightarrow{1.6} (M, 1.6, \mathbf{0}) \triangleleft p_1 \xrightarrow{b} (N, 1.6, [x = 1.6]) \triangleleft p_2 \xrightarrow{c} \text{no} \triangleleft p_3 \xrightarrow{4} \text{no} \triangleleft p_4 \xrightarrow{b} \dots$$

- ▶ Over $\text{ACT} = \{b, c\}$, M intuitively monitors for the formula $x \text{ in } \max(X, \langle x \in [1, 3] \rangle \langle b \rangle X \vee \langle x \notin [1, 3] \rangle \text{tt})$.
- ▶ Informally: If the first action occurs with a timestamp in $[1, 3]$, then every action within the absolute time interval $[1, 3]$ must be a b .

Reactive monitors

What about $\text{ACT} = \{b, c, c'\}$? Let

$$M = \text{LET } x \text{ IN } \underbrace{\text{rec } X.((x \in [1, 3])(\mathbf{b}.X + \mathbf{c}.\text{NO}) + (x \notin [1, 3]).\text{YES})}_N$$

and $p \xrightarrow{1.6} p_1 \xrightarrow{b} p_2 \xrightarrow{c'} p_3 \xrightarrow{4} p_4 \xrightarrow{b} p_5 \dots$ Then we have

$$m_0 \triangleleft p \xrightarrow{1.6} (M, 1.6, \mathbf{0}) \triangleleft p_1 \xrightarrow{b} (N, 1.6, [x = 1.6]) \triangleleft p_2 \xrightarrow{c'} \text{end} \triangleleft p_3 \xrightarrow{4} \text{end} \triangleleft p_4 \xrightarrow{b} \dots$$

M is now not reactive, it does not know what to say!

Since we are in a linear-time setting, we can:

- ▶ consider only reactive monitors (without loss of generality),
- ▶ abstract away the monitored program into the set of runs it produces
 - ▶ we write accordingly $m \xrightarrow{\theta} m'$ instead of $m \triangleleft p \xrightarrow{\theta} m' \triangleleft p'$ for some p and p' ,
 - ▶ we reason on all finite weak runs in $\mathcal{L}^*$ (and consequently on all weak runs in $\mathcal{L}^\omega$).

Monitorability: setting

Let $\text{TACT}_{div}^{\omega} \in \text{TACT}^{\omega}$ be the set of *time-divergent traces*

- ▶ For every $\pi \in \text{TACT}^{\omega}$ and every positive real r , there exists a position in π with timestamp greater than r .
- ▶ This assumption is natural, since in reality time always diverges.
 - ▶ example: the trace $(a_1, \frac{1}{2})(a_2, \frac{3}{4})(a_3, \frac{7}{8})(a_4, \frac{15}{16}) \dots$ is not realistic (global time bounded by 1),
 - ▶ no other assumption is made: actions can happen at the same time and delays may decrease arbitrarily as long as time eventually diverges,
 - ▶ exemple $(a_1, 0)(a_2, 0) \dots (a_6, \frac{1}{2})(a_7, \frac{3}{4})(a_8, \frac{7}{8}) \dots (a_{20}, 1)(a_{21}, 2) \dots$ is in $\text{TACT}_{div}^{\omega}$.
- ▶ Crucial for completeness of monitors.

We define the set of prefixes of a trace $\pi \in \text{TACT}^{\omega}$

$$Pf(\pi) = \{\pi' \in \text{TACT}^* \mid \exists \pi'' \in \text{TACT}^{\omega} : \pi' \cdot \pi'' = \pi\}$$

and infinite extensions of a finite trace $\pi \in \text{TACT}^*$

$$Ext(\pi) = \{\pi' \in \text{TACT}^{\omega} \mid \pi \in Pf(\pi')\}$$

Monitorability: monitor soundness & completeness

Definition (Monitor Soundness)

A monitor M is sound for $\phi \in T^{lin}$ if for every finite trace $\pi \in \text{TACT}^*$:

- ▶ $m_0 \xrightarrow{\text{TRN}(\pi)} \text{yes}$ implies $\forall \pi' \in \text{Ext}(\pi) : \pi' \in \llbracket \phi \rrbracket$
- ▶ $m_0 \xrightarrow{\text{TRN}(\pi)} \text{no}$ implies $\forall \pi' \in \text{Ext}(\pi) : \pi' \notin \llbracket \phi \rrbracket$

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Soundness is not sufficient, e.g., END is sound for every formula.

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Soundness is not sufficient, e.g., END is sound for every formula.

Definition (Strong completeness)

- ▶ M is satisfaction complete for ϕ if for all $\pi \in \text{TACT}_{div}^\omega$:

$$\pi \in \llbracket \phi \rrbracket \text{ implies } \exists \pi' \in \text{Pf}(\pi) : m_0 \xrightarrow{\text{TRN}(\pi')} \text{yes}$$

- ▶ M is violation complete for ϕ if for all $\pi \in \text{TACT}_{div}^\omega$:

$$\pi \notin \llbracket \phi \rrbracket \text{ implies } \exists \pi' \in \text{Pf}(\pi) : m_0 \xrightarrow{\text{TRN}(\pi')} \text{no}$$

- ▶ M is complete for ϕ if it is both violation and satisfaction complete for ϕ .

Monitorability

A formula $\phi \in \mathcal{T}^{lin}$ is complete (resp. violation, satisfaction) monitorable if there exists a monitor that is sound and complete (resp. violation complete, satisfaction complete) for it.

Problem

What is the largest fragment of $\mathcal{T}^{lin}$ such that every formula is complete (resp. violation, satisfaction) monitorable ?

Sound monitor synthesis

$\langle \text{ff} \rangle$	$=$	NO
$\langle \text{tt} \rangle$	$=$	YES
$\langle X \rangle$	$=$	X
$\langle \langle a \rangle \phi \rangle$	$=$	$a.\langle \phi \rangle + \bar{a}.\text{NO}$
$\langle \langle g \rangle \phi \rangle$	$=$	$g.\langle \phi \rangle + \bar{g}.\text{NO}$
$\langle x \text{ in } \phi \rangle$	$=$	LET x IN $\langle \phi \rangle$
$\langle \phi \wedge \psi \rangle$	$=$	$\langle \phi \rangle \otimes \langle \psi \rangle$
$\langle \phi \vee \psi \rangle$	$=$	$\langle \phi \rangle \oplus \langle \psi \rangle$
$\langle \max(X, \phi) \rangle$	$=$	$\langle \min(X, \phi) \rangle = \text{rec } X.\langle \phi \rangle$

- ▶ The compiler $\langle - \rangle$ is rather intuitive and straightforward.
- ▶ For example, for $\langle a \rangle \phi$, we want a monitor that monitors for ϕ if the action is a and reaches verdict NO otherwise.
- ▶ For $\langle \phi \wedge \psi \rangle$, we want a parallel monitor that reaches verdict YES iff both its submonitors reach verdict YES.
 - ▶ by definition of monitors, $M \otimes N$ is a YES verdict iff both M and N are YES verdicts.

Theorem

$\langle \phi \rangle$ is reactive, deterministic and sound for ϕ .

Largest complete-monitorable fragment?

- ▶ In the untimed case, the *recursion-free* fragment is its largest complete-monitorable fragment.
- ▶ This result does not hold in the timed setting.

Consider $\phi = \max(X, \langle \mathbf{b} \rangle X)$ meaning ‘globally $\mathbf{b}$ ’.

- ▶ $\langle \phi \rangle = \text{rec } X.(b.X + \bar{\mathbf{b}}.\text{NO})$ is sound and violation complete for ϕ .
- ▶ There is no sound and satisfaction-complete monitor for ϕ ...

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Consider now ϕ_c a *time-constrained* version of ϕ , i.e., using a *bounded* interval $I = [2, 5]$:

$$\phi_c = x \text{ \underline{in} } \max(X, (y \text{ \underline{in} } [y - x \in [2, 5]] \langle \mathbf{b} \rangle \text{tt}) \wedge \langle \text{ACT} \rangle X)$$

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Consider $\phi = \max(X, \langle \mathbf{b} \rangle X)$ meaning ‘globally **b**’.

- ▶ $\langle \phi \rangle = \text{rec } X.(b.X + \bar{\mathbf{b}}.\text{NO})$ is sound and violation complete for ϕ .
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Meaning “within the interval $[2, 5]$ (relative to the timestamp of the first action), every action must be a **b**”.

Here, a monitor can safely say YES when the time elapsed since the first action exceeds 5.

ϕ_c is therefore complete-monitorable.

Another example: constrained bounded response

- Formula ψ_c below expresses the constrained bounded response property, where I and I' are both bounded:
 “every action b occurring within interval I (relative to the first action) is followed by a c occurrence within I' (relative to such occurrence of b)”

$$\psi_c = x \underline{\text{in}} \max(Y, (y \underline{\text{in}} [y - x \in I][b]\zeta) \wedge (\langle \text{ACT} \rangle Y))$$

with:

$$\zeta = x' \underline{\text{in}} \min(X, (y' \underline{\text{in}} \langle y' - x' \in I' \rangle \langle c \rangle \text{tt}) \vee (\langle \text{ACT} \rangle X))$$

ψ_c is complete monitorable.

Trick to capture the largest complete-monitorable fragment

$$\psi_c = x \text{ \textbf{in} } \max(Y, (y \text{ \textbf{in} } [y - x \in I][b]\zeta) \wedge (z \text{ \textbf{in} } [z - x \prec \mathcal{RB}(I)]\langle \text{ACT} \rangle Y))$$

with:

$$\zeta = x' \text{ \textbf{in} } \min(X, (y' \text{ \textbf{in} } [y' - x' \in I']\langle c \rangle \text{tt}) \vee (z' \text{ \textbf{in} } [z' - x' \prec \mathcal{RB}(I')]\langle \text{ACT} \rangle X))$$

- ▶ with $\prec = <$ if the interval is right open and $\leq$ otherwise and $\mathcal{RB}(-)$ returns the right bound of its argument.
- ▶ We capture the complete-monitorable formulae by constraining recursion variables in time.

The complete-monitorable fragment of T^{lin}

$$\begin{array}{lcl}
 \phi, \psi & ::= & \texttt{tt} \quad | \quad \texttt{ff} \quad | \quad \phi \vee \psi \quad | \quad \phi \wedge \psi \\
 & & | \quad x \text{ \underline{in} } \phi \quad | \quad \langle a \rangle \phi \quad | \quad \langle g \rangle \phi \\
 & & | \quad \max(X, \phi) \quad | \quad \min(X, \phi) \quad | \quad x \text{ \underline{in} } \langle g_x \rangle \langle a \rangle X \\
 g_x & ::= & x \prec n \quad | \quad x - y \prec n \quad | \quad g_x \wedge g
 \end{array}$$

Definition (CT^{lin})

$\phi \in CT^{lin}$ iff ϕ is a formula generated by the above grammar, and for every $\max(X, \psi)$, $\min(X, \psi)$ subterms of ϕ , y is free in ψ .

Lemma

- ▶ For all $\max(X, \phi) \in CT^{lin}$, $\llbracket \max(X, \phi) \rrbracket = \llbracket \min(X, \phi) \rrbracket$
- ▶ For all $\min(X, \phi) \in CT^{lin}$, $\llbracket \min(X, \phi) \rrbracket = \llbracket \max(X, \phi) \rrbracket$

Theorem

For all $\phi \in CT^{lin}$, $\llbracket \phi \rrbracket$ is sound and strongly complete for ϕ .

Theorem

CT^{lin} is the largest complete-monitorable fragment of T^{lin} .

VT^{lin} : the violation monitorable fragment of T^{lin}

- Intuition : “relax” the syntax of the complete-monitorable fragment to allow general **max** formulae.

$$\begin{array}{lcl}
 \phi, \psi & ::= & \text{tt} \quad \quad \quad | \quad \text{ff} \quad \quad \quad | \quad \phi \vee \psi \quad \quad | \quad \phi \wedge \psi \\
 & | & x \text{ in } \phi \quad \quad \quad | \quad \langle a \rangle \phi \quad \quad \quad | \quad \langle g \rangle \phi \\
 & | & \max(X, \phi) \quad \quad \quad | \quad X \\
 & | & \cancel{\min(X, \phi)} \quad \quad \quad | \quad \cancel{x \text{ in } \langle g_x \rangle \langle a \rangle X}
 \end{array}$$

- example: $\phi_{\text{even}(b), [0, \infty)} = \max(X, \langle \text{ACT} \rangle \langle b \rangle X)$ (not expressible in MTL) is violation monitorable.

Theorem

VT^{lin} is the largest violation-monitorable fragment of T^{lin} .

VT^{lin} : another example

(Unconstrained) bounded response for any bounded I

$$\max(Y, ([b]\zeta \wedge \langle \text{ACT} \rangle Y))$$

with:

$$\zeta = x \text{ \underline{in} } \min(X, (y \text{ \underline{in} } \langle y - x \in I \rangle \langle c \rangle \text{tt}) \vee (\langle \text{ACT} \rangle X))$$

is violation monitorable.

ST^{lin} : the satisfaction-monitorable fragment of T^{lin}

- Intuition : “relax” the syntax of the complete-monitorable fragment to allow general **min** formulae.

$$\begin{array}{lcl}
 \phi, \psi & ::= & \texttt{tt} \quad \quad \quad | \texttt{ff} \quad \quad \quad | \phi \vee \psi \quad \quad | \phi \wedge \psi \\
 & | & x \underline{\text{in}} \phi \quad \quad \quad | \langle a \rangle \phi \quad \quad \quad | \langle g \rangle \phi \\
 & | & \min(X, \phi) \quad \quad \quad | X \\
 & | & \cancel{\max(X, \phi)} \quad \quad \quad | \cancel{x \text{ in } \langle g_x \rangle \langle a \rangle X}
 \end{array}$$

- example: $\phi = x \underline{\text{in}} \min(X, (\langle b \rangle y \underline{\text{in}} \langle y - x \in (0, 2] \rangle \langle c \rangle \texttt{tt}) \vee \langle \text{ACT} \rangle X)$
expressing the property:

“there exists a **b** immediately followed by a **c** within $(0, 2]$ ”
is satisfaction monitorable.

Theorem

ST^{lin} is the largest satisfaction-monitorable fragment of T^{lin}

Bounded intervals and monitorability

Remark

Bounded interval does not imply monitorable

Let $\phi = G (c \implies F_{[2,5]} (b \wedge F c))$. ϕ is not monitorable.

- ▶ For violations, take $\pi = (c, 1)(b, 4)(b, 8)(b, 9)(b, 10) \dots$. π is not in $\llbracket \phi \rrbracket$.
- ▶ Yet, there is no way for a monitor to safely say NO based on a finite prefix of $\pi \dots$
 - ▶ ... otherwise such monitor would be unsound.
- ▶ For satisfaction, take $\pi' = (b, 1)(b, 2)(b, 3) \dots$. π' is in $\llbracket \phi \rrbracket$.
- ▶ Yet, there is no way for a monitor to safely say YES based on a finite prefix of $\pi \dots$
 - ▶ ... otherwise such monitor would be unsound.

Conclusion

This work is the first that:

- ▶ proposes a timed μ -calculus (T^{lin}) strictly more expressive than MTL.
 - ▶ with complete proofs on monotonicity, continuity and expressiveness.
- ▶ solves the monitorability problem for such an expressive timed logic.
 - ▶ with complete proofs using e.g. a precise characterisation of unique fixed points,
 - ▶ and fully automated synthesis from MTL to T^{lin} and from T^{lin} to monitors.